CSE 484 MID-PROJECT · MEDSCRIBE

RESEARCH QUESTION

Do Doctor / Patient speaker labels make an LLM write a better SOAP note from a consultation transcript?

FIRST ANSWER: NO, NOT ON AVERAGE

ROUGE-L 0.316 with labels vs 0.317 without

3 exploratory train dialogues, Qwen2.5-7B, same prompt. Labels fixed a negation error but made the model echo the longest transcript: errors moved, they did not disappear.

CHANGED SINCE THE PROPOSAL

Old: build a 5-stage scribe (ASR, roles, NER, note, demo), no single question.

Why: feedback asked us to narrow it; NER, fine-tuning and the Turkish set are dropped.

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SYSTEM AND WHAT IS IMPLEMENTED

Four modules, one shared JSON format

1 Speech → text

WhisperX + pyannote

Alp · ASR lead · pretrained, no training

● implemented

→

2 Role assignment

TF-IDF + logistic regression

Yusuf · roles lead · trained by us

● implemented

→

3 SOAP note

Qwen2.5-7B 4-bit + frozen prompt

Ali · notes lead · pretrained, prompting

● implemented

→

4 Demo

Gradio, runs on one Colab T4

Akif · demo lead · interface

● text input works

EVERY MODULE READS AND WRITES THIS

```
{"dialogue_id": "aci_002", "source": "aci-bench",
 "segments": [{"id": 4, "speaker": "SPEAKER_00",
   "gold_role": "Doctor", "role": "Doctor", "role_prob": 0.91,
   "text": "have you noticed any rashes?"}, ...]}
```

DEMO STATUS

Pasted A/B text → roles → SOAP note works end to end.

Audio input is not connected yet.

Modules plug in by file name; mocks kept the demo running before they existed.

DATA AS LOADED

Published ACI-Bench splits, frozen test set

ACI-Bench (CC BY 4.0)	encounters	speakers	utterances	used for
train · train.csv	67	134	3,733	role model training
dev · valid.csv	20	40	1,012	development check
test · taskB_test1.csv	**40**	**80**	**2,032**	**frozen, run once**

FROZEN

Test labels are not used for training, prompt changes or model choice.

Preprocessing: [doctor]/[patient] tags → roles; [patient_guest] lines dropped (38 dev, 50 test). No other filtering.

PriMock57 (ASR only): 20 consultations, first 5 min, 2,050 utterances; gold speaker from channel energy.

ONE REAL EXAMPLE · aci_002 (train D2N053), transcript and reference note

```
[doctor] have you seen ever noticed any rashes either when it
  first started or intermittently?
[patient] on my body no not really
```

Reference note: "… Denies any rashes …"

Target: the note must keep the patient's denial.

STEP 1 · SPEECH → TEXT

Who spoke when: WhisperX + pyannote

93.8%
speaker accuracy, 1,923 / 2,050 utterances, 20 PriMock57 consultations

96.7%
on utterances longer than 1 second

87.8%
on short utterances (< 1 s)

≈
Whisper small vs medium model comparison: 93.8% vs 94.1%

SETUP

PriMock57 mock consultations (ACL 2022)

Doctor and patient channels mixed into one track

Whisper small + alignment, pyannote community-1 diarization, no training

Spread over consultations: 93.7 ± 5.0 pp (min 80.9%)

OBSERVATIONS

Errors cluster on short turns

One person reading both roles → found as one speaker

SPEAKER_00 / 01 numbers carry no role information, so the role model (next step) decides who is the doctor

Limitations: clean mock audio; gold labels from channel energy (approximate); speaker accuracy only, not WER/DER.

STEP 2 · ROLE ASSIGNMENT

Who is the doctor — without trusting A/B?

Speaker-level TF–IDF + balanced Logistic Regression · English clinical dialogue

01 **Group** Join every utterance from the same speaker.

02 **Vectorise** Word 1–2 grams + character 3–5 grams.

03 **Classify** Estimate P(Doctor) from language, not speaker ID.

04 **Constrain** For 2 speakers, assign exactly 1 Doctor + 1 Patient.

TRAINING DATA & FEATURES

67	ACI-Bench encounters
134	speaker documents · 67 Doctor + 67 Patient
9,782	word TF–IDF features
16,751	character TF–IDF features
26,533	total sparse features

0 speaker-name features · 0 GPU required

2 roles

1.11 MiB model

2.15 ms median inference

2,000 max iterations

assign_roles(dialogue) → role + P(Doctor) · MedScribe

BASELINE RESULTS · FROZEN TEST SET

Role assignment: near ceiling on clean transcripts

System (speaker level)	dev (20 enc.)	test (40 enc.) [95% CI]
First speaker is the doctor	100.0%	95.0% [87.5–100.0]
More “you / your” is the doctor	95.0%	97.5% [92.5–100.0]
TF-IDF + LR on speaker documents, ours	**95.0%**	**97.5% [92.5–100.0]**
Ablation: per-utterance model, mean per speaker	95.0%	97.5% [92.5–100.0]

1 / 40

test encounter wrong for our model (D2N114); the “you” rule makes the same mistake

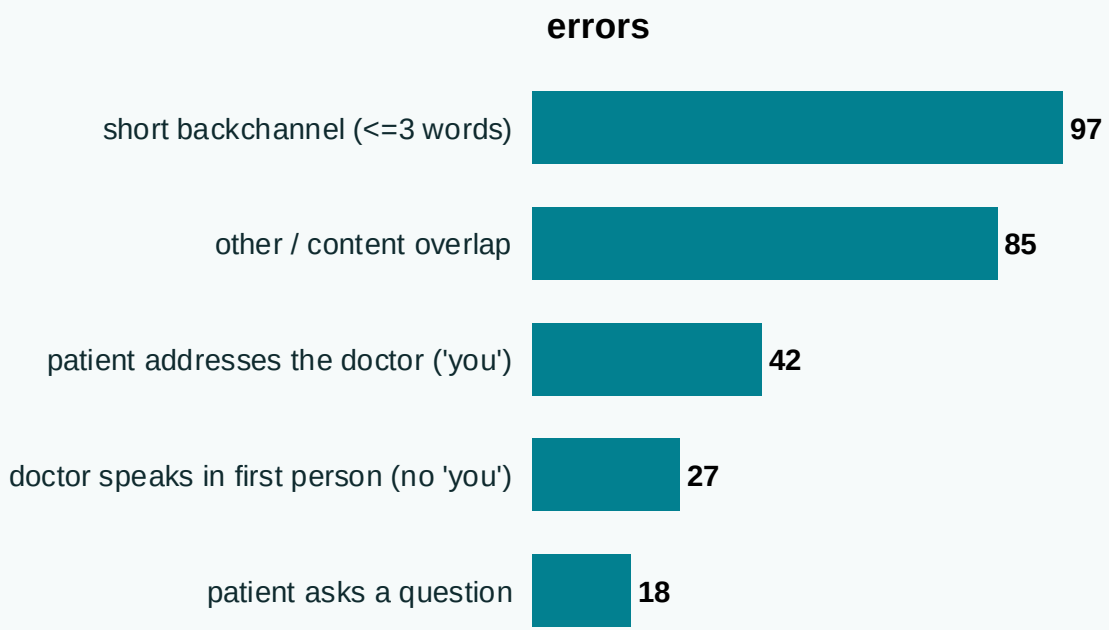
Spread: 95% bootstrap over test encounters (2,000 resamples, seed 0). On train, 5-fold grouped CV gives 92.5 ± 4.9. Ablation in its own unit: 86.8% [82.7–90.4] of 2,032 single utterances (encounter-cluster bootstrap); not comparable with the rows above.

WHAT THIS MEANS

On clean ACI-Bench transcripts our model ties a one-line rule. Single utterances are wrong 13% of the time, yet deciding per speaker gets 39/40 encounters right: the decision unit matters. Not yet tested: diarized ASR output, where speaker attribution is often wrong for short turns.

ERROR ANALYSIS · TEST SET · PER-UTTERANCE ABLATION MODEL

Per-utterance ablation: 269 errors, grouped



269 of 2,032 test utterances (13.2%), rule-based categories. The system's speaker-level model misses 1 of 40 encounters.

ONE EXAMPLE PER CATEGORY

short backchannel: “okay .”

other / content overlap: “u- u- um , maybe not necessarily . maybe in a , uh , few months we'll check o…”

patient addresses the doctor: “yeah , it's good seeing you too dr. hughes . you look like you're doing well .”

doctor speaks in first person: “i'm doing well , i'm doing well .”

patient asks a question: “i'm doing well . how are you ?”

WHAT THEY SUGGEST

Backchannels (“okay.”) carry no role signal → never decide per utterance, pool per speaker. Patients also say “you” and ask questions → the “you” rule and question counts break on short or role-reversed dialogues.

STEP 3 · NOTE GENERATION

SOAP notes from a transcript

Qwen2.5-7B-Instruct · 4-bit · temperature 0 · strict JSON output

01 **Data** 3 ACI-Bench dialogues → shared JSON; 25–26 utterances, check_format clean.

02 **Transcript** With roles the lines read "Doctor: … / Patient: …"; without roles, plain text only.

03 **Prompt** One frozen prompt; the model may use only what the transcript states.

04 **Output** JSON keyed S / O / A / P, so each section can be checked on its own.

TWO RULES FROM THE FROZEN PROMPT

> “Negation matters. If the patient denies a symptom, record it as denied — never as present.”
>
> **Negation**

> “Do not turn the doctor’s questions into patient complaints. A question is not a symptom.”
>
> **Question ≠ complaint**

3 dialogues

2 conditions

6 notes

0 training

METHOD STATUS · FIRST NUMBERS (EXPLORATORY)

With roles vs without: same model, same prompt

ROUGE vs reference note	R-1	R-2	R-L
Without labels (baseline)	0.481	0.212	0.317
With Doctor/Patient labels	0.479	0.217	0.316

Mean of 3 exploratory dialogues from the train split (D2N050, D2N053, D2N060), not the dev set. Qwen2.5-7B-Instruct 4-bit, temperature 0, frozen prompt v1.

ONE CONTROLLED CHANGE

Only the transcript changes: “Doctor: … / Patient: …” vs plain lines in the same order.

Prompt, model, decoding and data are identical.

ROUGE barely moves (≤ 0.005), so we read every note side by side →

WHAT THE 6 NOTES SHOWED

With labels
1 of 3 notes echoed the transcript (aci_003: 3,229 chars, 7 of 9 S sentences in first person)

Without labels
1 of 3 lost negation scope (aci_002: “no joint pain, fever, weight loss”)

Both
118/72 written as 180/72; “push fluids” added though nobody said it

PLAN TO THE FINAL

Remaining experiments, owners and dates

Experiment	Owner	Date
Run with/without labels on the 40 frozen test dialogues; ROUGE + BERTScore	Notes lead	Nov 26
Ablation: anonymous A/B labels (turns, no roles) as a 3rd condition	Notes lead	Nov 26
Attribution-error checklist; 2 people label 10 notes for agreement	Demo + notes	Dec 3
BERT role classifier vs TF-IDF on the same frozen test	Roles lead	Nov 26
Hand-check 30 utterance errors; roles on diarized ASR output	Roles + ASR	Dec 3
Predicted vs gold roles into the note (error propagation)	ASR + notes	Dec 10
Audio input in the demo; report skeleton	Demo lead	Dec 10

ABLATION

No speaker info → anonymous A/B → Doctor/Patient. Separates “knowing the turns” from “knowing the roles”.

BIGGEST RISK

ROUGE cannot see who-said-what errors, so a real effect looks like noise.

Fallback: count negation flips and question→complaint errors with a fixed checklist.

1 Title and abstract

MedScribe: Do speaker roles help an LLM write clinical notes?

Ambient scribes turn a doctor–patient conversation into a SOAP note, but diarization only gives anonymous speakers. We train a text-based role classifier (TF-IDF + logistic regression; 97.5% speaker accuracy on the frozen ACI-Bench test) and test whether Doctor/Patient labels change the note an open LLM (Qwen2.5-7B) writes from the same transcript. On three exploratory train dialogues, labels did not change ROUGE (R-L 0.316 vs 0.317) but shifted the error types. On the 40-dialogue test set, labels [RESULT: ROUGE / BERTScore / attribution errors, with vs without vs anonymous A/B]. (≈120 words)

2 Research question

Do Doctor/Patient speaker labels make an LLM write a better SOAP note? First result: no, not on average (3 exploratory train dialogues). Changed: the proposal was a 5-stage system with no single question; narrowed after feedback (NER, fine-tuning, Turkish data dropped).

3 Contributions (testable claims)

C1. Labels reduce attribution errors (negation flips, questions written as complaints) on the 40 test dialogues, even when ROUGE is unchanged.

C2. Most of any gain comes from turn structure: anonymous A/B labels close at least half of the gap between no speaker info and roles.

C3. Role decisions must be made per speaker: the per-utterance model is wrong on 13% of test utterances, yet averaging its scores per speaker gets 97.5% of encounters right.

4 Related work (how we differ)

Krishna et al., ACL-IJCNLP 2021 — Generating SOAP notes from doctor–patient conversations using modular summarization. Fine-tuned extract-then-abstract; we prompt an open LLM and isolate role labels.

Yim et al., Scientific Data 2023 — ACI-BENCH. Benchmark we use; we add a role-label ablation on it.

Ben Abacha et al., EACL 2023 — An empirical study of clinical note generation (MTS-Dialog). Section generation; we study one input factor, speaker roles.

Ben Abacha et al., ClinicalNLP 2023 — Overview of MEDIQA-Chat 2023. Shared task systems; none separates the effect of role tags.

Korfiatis et al., ACL 2022 — PriMock57. Audio corpus we use only for speaker separation.

Bain et al., Interspeech 2023 — WhisperX; Bredin et al., ICASSP 2020 — pyannote. Tools we use as is; we solve the role mapping they leave open.

5 Data

ACI-Bench, CC BY 4.0. As loaded: train 67 enc. / 3,733 utt.; dev 20 / 1,012; test (clinicalnlp_taskB_test1, frozen) 40 / 2,032.

Preprocessing: [patient_guest] lines dropped (dev 38, test 50); encounters unchanged.

PriMock57, CC BY 4.0: 20 consultations, first 5 min, 2,050 utterances. Example: [doctor] have you … noticed any rashes? / [patient] on my body no not really.

No data labelled by us yet; the attribution-error checklist will report agreement on 10 notes.

6 Method and baseline

Baseline: note from the transcript without speaker labels. Method: same transcript with Doctor/Patient labels from our classifier (gold labels for the controlled run). The one change: the labels.

Compute used: 1 Colab T4, ~1–2 min per note (4-bit 7B); roles: 2.15 ms warm inference, ~1–2.6 s with Python start-up and model loading (tested Mac). Needed: ~4 T4-hours for 40 dialogues × 3 conditions.

7 Evaluation protocol (fixed before method results)

Metrics: ROUGE-1/2/L, BERTScore vs reference note; attribution-error count per note (checklist). Test: ACI-Bench taskB_test1, 40 dialogues, frozen. Temperature 0 (one run); spread by paired bootstrap over dialogues (2,000, seed 0). Real difference = paired 95% CI excludes 0.

8 Results so far (final table)

System	Roles: speaker acc. (test)	Note R-L (train, n=3)	Note R-L (test, n=40)	Attribution errors
First speaker rule	95.0% [87.5–100.0]	–	–	–
TF-IDF + LR roles (ours)	97.5% [92.5–100.0]	–	–	–
Note, no speaker info (baseline)	–	0.317		
Note, anonymous A/B	–			
Note, Doctor/Patient labels	–	0.316		

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Error analysis

Per-utterance ablation model (not the system's speaker-level model, which misses 1/40): 269 test errors, grouped: short backchannel 97; other / content overlap 85; patient addresses the doctor 42; doctor speaks in first person 27; patient asks a question 18. Example: “okay.” (short backchannel). Notes: 1 echo, 1 negation-scope, 1 number, 1 added fact (6 notes).

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Research lead section

Research lead: the notes and experiment lead (slide 2), who owns the research question. Two recent papers: ACI-BENCH (Yim et al. 2023) and MEDIQA-Chat 2023 (Ben Abacha et al.); both report note quality without isolating speaker labels. We run the ablation no-speaker / anonymous A/B / roles.

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Plan and risk

Run with/without labels on the 40 frozen test dialogues; ROUGE + BERTScore — Notes lead, Nov 26; Ablation: anonymous A/B labels (turns, no roles) as a 3rd condition — Notes lead, Nov 26; Attribution-error checklist; 2 people label 10 notes for agreement — Demo + notes, Dec 3; BERT role classifier vs TF-IDF on the same frozen test — Roles lead, Nov 26; …(full list on slide 10). Risk: ROUGE blind to attribution errors → fallback checklist. Feedback response: scope narrowed to one question.

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Reproducibility

Repo: github.com/Pikseel/medscribe. Field 8: python -m eval.role_eval (roles) and python -m notes.generate_note --experiment (notes, needs GPU). Seeds: bootstrap 0, LLM temperature 0. Versions: Python 3.12, scikit-learn ≥1.4, transformers + bitsandbytes (Colab), WhisperX.

Who did what: one module per member, as on slide 2 — ASR lead: speech → text; roles lead: role classifier; notes lead: notes and experiment; demo lead: shared format, demo, evaluation, slides. AI tools: Claude (Anthropic) for planning, code scaffolding, evaluation scripts and slide drafts; all outputs checked by the team.